

ALFAA33266

## Phosphoric acid, 85% w/w aqueous solution, ACS

### SECTION 1. IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

**产品说明:**  
**Product Description:** 正磷酸  
Phosphoric acid, 85% w/w aqueous solution, ACS

**Cat No. :** 33266  
**Synonyms** Orthophosphoric acid  
**CAS No** 7664-38-2  
**Molecular Formula** H3 O4 P

**Supplier** Avocado Research Chemicals Ltd.  
(Part of Thermo Fisher Scientific)  
Shore Road, Heysham  
Lancashire, LA3 2XY,  
United Kingdom  
Office Tel: +44 (0) 1524 850506  
Office Fax: +44 (0) 1524 850608

**Emergency Telephone Number** For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

**E-mail address** begel.sdsdesk@thermofisher.com

**Recommended Use** Laboratory chemicals.  
**Uses advised against** No Information available

### SECTION 2. HAZARD IDENTIFICATION

**Physical State**  
Liquid

**Appearance**  
Clear, Viscous

**Odor**  
Odorless

#### Emergency Overview

Causes severe skin burns and eye damage. May be corrosive to metals. Harmful if swallowed. May be harmful in contact with skin.  
Hygroscopic.

#### Classification of the substance or mixture

|                                        |              |
|----------------------------------------|--------------|
| Substances/mixtures corrosive to metal | Category 1   |
| Acute Oral Toxicity                    | Category 4   |
| Acute Dermal Toxicity                  | Category 5   |
| Skin Corrosion/Irritation              | Category 1 B |
| Serious Eye Damage/Eye Irritation      | Category 1   |

#### Label Elements



# SAFETY DATA SHEET

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**Signal Word****Danger****Hazard Statements**

H290 - May be corrosive to metals  
H314 - Causes severe skin burns and eye damage  
H302 - Harmful if swallowed  
H313 - May be harmful in contact with skin

**Precautionary Statements****Prevention**

P234 - Keep only in original packaging  
P264 - Wash face, hands and any exposed skin thoroughly after handling  
P270 - Do not eat, drink or smoke when using this product  
P271 - Use only outdoors or in a well-ventilated area  
P280 - Wear protective gloves/protective clothing/eye protection/face protection

**Response**

P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower  
P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing  
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
P310 - Immediately call a POISON CENTER or doctor  
P330 - Rinse mouth  
P331 - Do NOT induce vomiting  
P390 - Absorb spillage to prevent material damage  
P362 + P364 - Take off contaminated clothing and wash it before reuse

**Storage**

P402 - Store in a dry place  
P403 + P233 - Store in a well-ventilated place. Keep container tightly closed  
P406 - Store in corrosion resistant polypropylene container with a resistant liner

**Disposal**

P501 - Dispose of contents/ container to an approved waste disposal plant

**Physical and Chemical Hazards**

May be corrosive to metals. Hygroscopic.

**Health Hazards**

Corrosive. Causes skin and eye burns. Causes serious eye damage. Harmful if swallowed. May be harmful in contact with skin.

**Environmental hazards**

Contains no substances known to be hazardous to the environment or not degradable in waste water treatment plants. Will likely be mobile in the environment due to its water solubility. The product is water soluble, and may spread in water systems.

**Other Hazards**

This product does not contain any known or suspected endocrine disruptors.

**SECTION 3. COMPOSITION/INFORMATION ON INGREDIENTS**

| Component            | CAS No    | Weight % |
|----------------------|-----------|----------|
| Orthophosphoric acid | 7664-38-2 | >= 85    |
| Water                | 7732-18-5 | <= 15    |

**SECTION 4. FIRST AID MEASURES****General Advice**

Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.

**Eye Contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Immediate medical attention is required.

**Skin Contact**

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Wash off immediately with plenty of water for at least 15 minutes. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Call a physician immediately.

**Inhalation**

If not breathing, give artificial respiration. Remove from exposure, lie down. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Call a physician immediately.

**Ingestion**

Do NOT induce vomiting. Clean mouth with water. Never give anything by mouth to an unconscious person. Call a physician immediately.

**Most important symptoms and effects**

Causes burns by all exposure routes. Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation: Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated

**Self-Protection of the First Aider**

Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

**Notes to Physician**

Treat symptomatically.

### SECTION 5. FIRE-FIGHTING MEASURES

**Suitable Extinguishing Media**

Substance is nonflammable; use agent most appropriate to extinguish surrounding fire. CO<sub>2</sub>, dry chemical, dry sand, alcohol-resistant foam.

**Extinguishing media which must not be used for safety reasons**

None.

**Specific Hazards Arising from the Chemical**

Thermal decomposition can lead to release of irritating gases and vapors. The product causes burns of eyes, skin and mucous membranes.

**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

### SECTION 6. ACCIDENTAL RELEASE MEASURES

**Personal Precautions**

Use personal protective equipment as required. Ensure adequate ventilation. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak.

**Environmental Precautions**

Should not be released into the environment.

**Methods for Containment and Clean Up**

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal.

Refer to protective measures listed in Sections 8 and 13.

### SECTION 7. HANDLING AND STORAGE

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**Handling**

Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Use only under a chemical fume hood. Do not breathe mist/vapors/spray. Do not ingest. If swallowed then seek immediate medical assistance.

**Storage**

Keep containers tightly closed in a dry, cool and well-ventilated place. Corrosives area. Store under an inert atmosphere. Protect from moisture.

**Specific Use(s)**

Use in laboratories

**SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION**
**Control Parameters**

| Component            | China                                                 | Taiwan                   | Thailand                 | Hong Kong                                             |
|----------------------|-------------------------------------------------------|--------------------------|--------------------------|-------------------------------------------------------|
| Orthophosphoric acid | TWA: 1 mg/m <sup>3</sup><br>STEL: 3 mg/m <sup>3</sup> | TWA: 1 mg/m <sup>3</sup> | TWA: 1 mg/m <sup>3</sup> | TWA: 1 mg/m <sup>3</sup><br>STEL: 3 mg/m <sup>3</sup> |

| Component            | ACGIH TLV                                             | OSHA PEL                                                                                              | NIOSH                                                                                 | The United Kingdom                                    | European Union                                                     |
|----------------------|-------------------------------------------------------|-------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|-------------------------------------------------------|--------------------------------------------------------------------|
| Orthophosphoric acid | TWA: 1 mg/m <sup>3</sup><br>STEL: 3 mg/m <sup>3</sup> | (Vacated) TWA: 1 mg/m <sup>3</sup><br>(Vacated) STEL: 3 mg/m <sup>3</sup><br>TWA: 1 mg/m <sup>3</sup> | IDLH: 1000 mg/m <sup>3</sup><br>TWA: 1 mg/m <sup>3</sup><br>STEL: 3 mg/m <sup>3</sup> | STEL: 2 mg/m <sup>3</sup><br>TWA: 1 mg/m <sup>3</sup> | TWA: 1 mg/m <sup>3</sup> (8h)<br>STEL: 2 mg/m <sup>3</sup> (15min) |

Legend

ACGIH - American Conference of Governmental Industrial Hygienists

OSHA - Occupational Safety and Health Administration

NIOSH: NIOSH - National Institute for Occupational Safety and Health

**Exposure Controls****Engineering Measures**

Use only under a chemical fume hood. Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source.

**Personal protective equipment**

**Eye Protection** Goggles (European standard - EN 166)

**Hand Protection** Protective gloves

| Glove material | Breakthrough time | Glove thickness | EU standard       | Glove comments                                                                 |
|----------------|-------------------|-----------------|-------------------|--------------------------------------------------------------------------------|
| Butyl rubber   | > 480 minutes     | 0.36 mm         | EN 374<br>Level 6 | As tested under EN374-3 Determination of Resistance to Permeation by Chemicals |
| Nitrile rubber | > 480 minutes     | 0.1 mm          |                   |                                                                                |
| Neoprene       | > 480 minutes     | 0.45 mm         |                   |                                                                                |
| Viton (R)      | > 480 minutes     | 0.7 mm          |                   |                                                                                |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

**Skin and body protection** Wear appropriate protective gloves and clothing to prevent skin exposure

**Respiratory Protection** When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.

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|                                        |                                                                                                                                                                                                                                                                                                                             |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                        | To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly                                                                                                                                                                                                         |
| <b>Large scale/emergency use</b>       | Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced<br><b>Recommended Filter type:</b> Particulates filter conforming to EN 143 Acid gases filter Type E Yellow conforming to EN14387                                       |
| <b>Small scale/Laboratory use</b>      | Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.<br><b>Recommended half mask:-</b> Valve filtering: EN405; or; Half mask: EN140; plus filter, EN 141<br>When RPE is used a face piece Fit Test should be conducted |
| <b>Hygiene Measures</b>                | Handle in accordance with good industrial hygiene and safety practice.                                                                                                                                                                                                                                                      |
| <b>Environmental exposure controls</b> | Prevent product from entering drains. Do not allow material to contaminate ground water system.                                                                                                                                                                                                                             |

## SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES

|                                                |                          |                                          |
|------------------------------------------------|--------------------------|------------------------------------------|
| <b>Appearance</b>                              | Clear, Viscous           |                                          |
| <b>Physical State</b>                          | Liquid                   |                                          |
| <b>Odor</b>                                    | Odorless                 |                                          |
| <b>Odor Threshold</b>                          | No data available        |                                          |
| <b>pH</b>                                      | < 2                      |                                          |
| <b>Melting Point/Range</b>                     | 21 °C / 69.8 °F          |                                          |
| <b>Softening Point</b>                         | No data available        |                                          |
| <b>Boiling Point/Range</b>                     | 158 °C / 316.4 °F        |                                          |
| <b>Flash Point</b>                             | No information available | <b>Method -</b> No information available |
| <b>Evaporation Rate</b>                        | Not applicable           |                                          |
| <b>Flammability (solid,gas)</b>                | Not applicable           | Liquid                                   |
| <b>Explosion Limits</b>                        | Not applicable           |                                          |
| <b>Vapor Pressure</b>                          | 2 hPa @ 20°C             |                                          |
| <b>Vapor Density</b>                           | 3.4                      | (Air = 1.0)                              |
| <b>Specific Gravity / Density</b>              | 1.680                    |                                          |
| <b>Bulk Density</b>                            | Not applicable           | Liquid                                   |
| <b>Water Solubility</b>                        | Miscible                 |                                          |
| <b>Solubility in other solvents</b>            | No information available |                                          |
| <b>Partition Coefficient (n-octanol/water)</b> |                          |                                          |
| <b>Autoignition Temperature</b>                | No data available        |                                          |
| <b>Decomposition Temperature</b>               | 300 °C                   |                                          |
| <b>Viscosity</b>                               | 32 mPas                  | 30°C                                     |
| <b>Explosive Properties</b>                    | Not applicable           |                                          |
| <b>Oxidizing Properties</b>                    | Not applicable           |                                          |
| <b>Molecular Formula</b>                       | H3 O4 P                  |                                          |
| <b>Molecular Weight</b>                        | 98.00                    |                                          |

## SECTION 10. STABILITY AND REACTIVITY

|                                 |                                          |
|---------------------------------|------------------------------------------|
| <b>Stability</b>                | Hygroscopic.                             |
| <b>Hazardous Reactions</b>      | None under normal processing.            |
| <b>Hazardous Polymerization</b> | Hazardous polymerization does not occur. |

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**Conditions to Avoid** Incompatible products. Excess heat. Exposure to moisture. Exposure to moist air or water.

**Materials to avoid** Strong oxidizing agents. Metals. Bases. Alcohols. Amines. halogenated agents.

**Hazardous Decomposition Products** Oxides of phosphorus.

### SECTION 11. TOXICOLOGICAL INFORMATION

#### Product Information

(a) acute toxicity;  
**Toxicology data for the components**

| Component            | LD50 Oral                 | LD50 Dermal                  | LC50 Inhalation                   |
|----------------------|---------------------------|------------------------------|-----------------------------------|
| Orthophosphoric acid | LD50 = 1530 mg/kg ( Rat ) | LD50 = 2740 mg/kg ( Rabbit ) | 850 mg/m <sup>3</sup> ( Rat ) 1 h |
| Water                | -                         | -                            | -                                 |

(b) skin corrosion/irritation; Category 1 B

(c) serious eye damage/irritation; Category 1

(d) respiratory or skin sensitization;  
 Respiratory No data available  
 Skin No data available

(e) germ cell mutagenicity; No data available

(f) carcinogenicity; No data available  
 There are no known carcinogenic chemicals in this product

(g) reproductive toxicity; No data available

(h) STOT-single exposure; No data available

(i) STOT-repeated exposure; No data available

Target Organs None known.

(j) aspiration hazard; Based on available data, the classification criteria are not met

**Symptoms / effects, both acute and delayed** Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation: Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated

### SECTION 12. ECOLOGICAL INFORMATION

**Ecotoxicity effects** Do not empty into drains. Large amounts will affect pH and harm aquatic organisms.

| Component            | Freshwater Fish         | Water Flea             | Freshwater Algae       | Microtox |
|----------------------|-------------------------|------------------------|------------------------|----------|
| Orthophosphoric acid | 98 - 106 mg/L LC50 96 h | > 100 mg/L EC50 = 48 h | > 100 mg/L EC50 = 72 h |          |

# SAFETY DATA SHEET

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**Persistence and Degradability****Persistence****Degradation in sewage treatment plant**

Miscible with water, Persistence is unlikely, based on information available.  
Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

**Bioaccumulative Potential**

Bioaccumulation is unlikely

**Mobility in soil**

The product is water soluble, and may spread in water systems Will likely be mobile in the environment due to its water solubility Highly mobile in soils

**Endocrine Disruptor Information**  
**Persistent Organic Pollutant**  
**Ozone Depletion Potential**

This product does not contain any known or suspected endocrine disruptors  
This product does not contain any known or suspected substance  
This product does not contain any known or suspected substance

**SECTION 13. DISPOSAL CONSIDERATIONS****Waste from Residues/Unused Products**

Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.

**Contaminated Packaging**

Dispose of this container to hazardous or special waste collection point.

**Other Information**

Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains. Do not flush to sewer. Large amounts will affect pH and harm aquatic organisms. Solutions with low pH-value must be neutralized before discharge.

**SECTION 14. TRANSPORT INFORMATION****Road and Rail Transport**

|                      |                           |
|----------------------|---------------------------|
| UN-No                | UN1805                    |
| Proper Shipping Name | PHOSPHORIC ACID, SOLUTION |
| Hazard Class         | 8                         |
| Packing Group        | III                       |

**IMDG/IMO**

|                      |                          |
|----------------------|--------------------------|
| UN-No                | UN1805                   |
| Proper Shipping Name | PHOSPHORIC ACID SOLUTION |
| Hazard Class         | 8                        |
| Packing Group        | III                      |

**IATA**

|                      |                           |
|----------------------|---------------------------|
| UN-No                | UN1805                    |
| Proper Shipping Name | PHOSPHORIC ACID, SOLUTION |
| Hazard Class         | 8                         |
| Packing Group        | III                       |

**Special Precautions for User** No special precautions required

**SECTION 15. REGULATORY INFORMATION****International Inventories**

X = listed, China (IECSC), Europe (EINECS/ELINCS/NLP), U.S.A. (TSCA), Canada (DSL/NDL), Philippines (PICCS), Japan (ENCS), Japan

# SAFETY DATA SHEET

## Phosphoric acid, 85% w/w aqueous solution, ACS

(ISHL), Australia (AICS), Korea (KECL).

| Component            | The Inventory of Hazardous Chemicals (2015 Edition) | List of dangerous goods GB 12268 - 2012 | TCSI | IECSC | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | AICS | KECL     |
|----------------------|-----------------------------------------------------|-----------------------------------------|------|-------|-----------|------|-----|-------|------|------|------|----------|
| Orthophosphoric acid | X                                                   | X                                       | X    | X     | 231-633-2 | X    | X   | X     | X    | X    | X    | KE-27427 |
| Water                | -                                                   | -                                       | X    | X     | 231-791-2 | X    | X   | X     | X    |      | X    | KE-35400 |

### National Regulations

## SECTION 16. OTHER INFORMATION

**Prepared By** Health, Safety and Environmental Department  
**Creation Date** 19-Oct-2009  
**Revision Date** 29-Apr-2024  
**Revision Summary** New emergency telephone response service provider.

### Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.  
 Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.  
 First aid for chemical exposure, including the use of eye wash and safety showers.  
 Chemical incident response training.

### Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**DNEL** - Derived No Effect Level

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**NOEC** - No Observed Effect Concentration

**PBT** - Persistent, Bioaccumulative, Toxic

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**PNEC** - Predicted No Effect Concentration

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (Volatile Organic Compound)

### Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

### Physical hazards

On basis of test data

### Health Hazards

Bridging principle "Dilution" Calculation method



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Environmental hazards

Bridging principle "Dilution" Calculation method

## Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**